

ALEXANDRIA, Romania

WMO: 154890

Lat: 43.9781N

Lon: 25.3528E

Elev: 77

StdP: 100.41

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.4	-13.8	1.1	-11.5	-11.0	1.5	-9.3	14.3	1.2	11.7	-0.3	1.6	270	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	34.7	21.6	33.2	20.9	31.8	20.5	23.1	31.5	22.4	30.2	21.7	29.1	2.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.5	25.6	20.0	14.8	25.2	19.2	14.1	24.4	68.9	31.5	66.1	30.5	63.4	29.2	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.4	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-1.7	3.1	7.6	11.7	17.5	22.3	23.9	24.2	19.2	12.2	5.8	1.3
	DBStd	9.59	4.91	4.78	4.41	4.17	3.14	3.00	2.53	2.79	3.71	4.40	4.53	4.14
	HDD10.0	1124	363	195	98	30	1	0	0	0	0	28	141	268
	HDD18.3	2766	621	426	332	201	52	4	0	2	34	191	377	527
	CDD10.0	1968	0	3	24	81	234	368	431	439	276	98	14	0
	CDD18.3	569	0	0	0	2	27	122	173	183	61	3	0	0
	CDH23.3	6094	0	0	5	55	308	1195	1812	2028	641	49	1	0
	CDH26.7	2576	0	0	0	8	65	486	807	978	226	7	0	0
Wind	WSAvg	2.7	2.7	3.0	3.1	3.3	2.8	2.4	2.4	2.6	2.7	2.6	2.5	2.7
Precipitation	PrecAvg	544	37	28	34	47	62	65	55	35	52	47	34	43
	PrecMax	865	104	80	79	94	137	147	177	132	148	108	83	152
	PrecMin	313	2	2	1	16	15	9	9	3	5	0	0	3
	PrecStd	124	23	18	20	21	27	34	39	30	35	32	22	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	18.5	23.9	27.8	30.9	35.9	37.0	36.7	33.8	28.1	21.2	13.3
		MCWB	7.9	13.1	14.2	17.5	20.7	22.0	23.5	21.7	20.1	17.6	14.6	9.6
	2%	DB	8.1	14.8	20.4	24.8	28.6	33.2	34.3	34.7	31.3	24.9	17.2	10.9
		MCWB	5.8	10.7	12.3	16.4	18.8	21.5	21.8	21.5	18.8	16.0	13.0	8.2
	5%	DB	6.0	12.1	18.0	21.8	26.9	31.2	32.8	33.2	29.1	22.4	14.3	8.4
		MCWB	4.4	9.0	11.1	14.4	18.1	21.3	21.1	20.7	17.9	15.0	11.5	6.5
	10%	DB	4.3	9.7	15.3	19.5	24.9	29.5	31.1	31.9	26.9	19.9	12.0	6.4
		MCWB	3.3	7.4	9.7	13.2	17.2	20.4	20.6	20.3	17.7	13.9	10.2	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	13.5	14.6	18.3	21.4	24.2	23.7	24.1	21.8	18.3	15.2	10.1
		MCDB	11.2	18.1	22.7	26.3	29.1	32.2	34.0	34.2	30.7	24.9	20.2	12.9
	2%	WB	6.0	11.0	12.9	16.9	20.0	23.0	23.0	23.0	20.0	16.8	13.6	8.5
		MCDB	7.7	14.2	19.3	24.4	26.5	30.2	31.4	31.8	27.4	22.4	17.2	10.9
	5%	WB	4.7	9.3	11.4	15.5	18.8	22.1	22.2	22.0	19.1	15.7	12.2	7.0
		MCDB	5.7	12.0	17.3	21.2	25.2	29.4	29.9	30.1	26.6	21.4	14.5	8.4
	10%	WB	3.3	7.7	10.0	14.0	17.8	21.3	21.6	21.2	18.3	14.6	10.8	5.4
		MCDB	4.1	9.7	14.8	18.5	23.7	28.0	28.7	29.3	25.5	19.2	12.1	6.4
Mean Daily Temperature Range	5% DB	MDBR	6.4	8.4	10.5	11.2	11.8	12.0	12.6	13.3	12.3	10.6	6.8	6.1
		MCDBR	9.0	12.8	15.8	15.9	14.8	14.4	15.0	15.5	15.7	15.3	11.1	9.3
		MCWBR	6.3	8.8	8.7	8.3	6.1	5.0	4.6	4.9	5.7	7.2	6.7	7.1
	5% WB	MCDBR	7.7	12.3	14.9	14.6	13.7	13.5	13.2	13.7	13.2	13.6	10.2	9.4
	MCWBR	5.9	8.6	8.3	7.7	6.2	5.3	4.6	5.0	5.1	6.8	6.0	7.2	
Clear-Sky Solar Irradiance	taub	0.347	0.379	0.415	0.462	0.437	0.446	0.456	0.470	0.435	0.403	0.375	0.352	
	taud	2.275	2.204	2.155	2.043	2.192	2.204	2.181	2.137	2.201	2.271	2.303	2.292	
	Ebn at Noon	752	791	811	802	838	830	815	787	783	756	714	705	
	Edh at Noon	87	111	132	160	142	141	143	144	125	103	85	79	
All-Sky Solar Radiation	RadAvg	1.45	2.26	3.44	4.74	5.87	6.42	6.49	5.79	4.20	2.73	1.54	1.18	
	RadStd	0.22	0.32	0.35	0.48	0.46	0.44	0.42	0.35	0.33	0.32	0.21	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-128	N/A	N/A

Nomenclature: See separate page